

Project Report : AI Powered Keyword Extraction from Problem Solving Images

GROUP 3

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Introduction

With the rapid advancement in AI technology in recent years, the idea of AI in education has frequently been discussed by some scholars [3][5], providing personalized education [2][5] which motivates students to improve their understanding on the subject despite being unsatisfied with the provided service [2]. AI however should not be treated as a separate entity that rivals educators since AI lacks the ability to build on top of student responses [1]; instead it should be treated as a tool to aid students[4]. Therefore in this project, we propose a new AI-driven feature that extracts important keywords from exam question images, to help students record and revisit unsolved questions. Through the development of a prototype and a user study, we found that our new feature has the potential to significantly enhance students' study habits.

Motivations

We derived inspiration for this project from an existing mobile app service in the KSAT(대학수학능력시험) preparation category. One member of this HCI project has been part of the management team of the app service, exploring users' needs to help them better prepare KSAT. Utilizing such experiences, we aimed to make a new helpful feature driven by AI, regarding KSAT preparation.

Among various aspects that students prepare for KSAT, we especially wanted to help students record and revisit unsolved questions after they take mock-tests. Here, 'unsolved questions' mean exam questions that were not solved during exam time due to lack of ability or lack of time. It was already known that many students have had difficulty recording unsolved questions, since in the existing app, a feature to record unsolved questions had been less-used by users. Therefore, we desired to build a new feature for easier recording and revisiting of unsolved questions.

Goal

We hoped to build a prototype to allow users to test our new feature. Next, we wanted to measure (1) **if the new feature is actually helpful for the betterment of users' study habits** and (2) **if the new feature would encourage users to record unsolved questions using the app.**

The existing app already included a feature to record and revisit unsolved questions, as in Figure 1. Thus our first goal was to develop a new AI-powered feature to extract important keywords from question images, which can be applied on the existing service. **We aimed to implement the prototype as a new website to test it safely, not adding the new feature in the existing app service directly.** After developing the prototype, we planned to conduct pre- and post- surveys from users to

evaluate the impact of the prototype. A paired t-test would be used to measure if there was statistically significant improvement in willingness of users after trying our new feature.

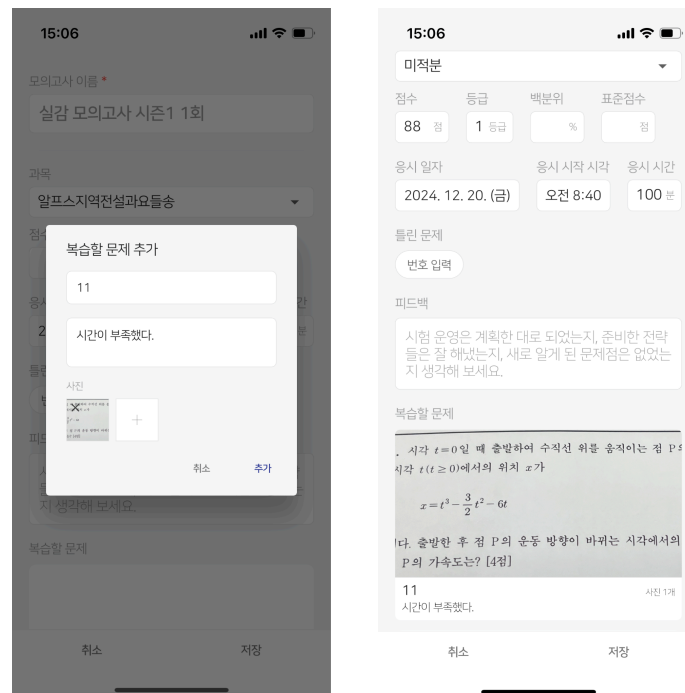


Figure 1 : Existing features in the existing app service, not including keyword extraction

HCI Methodology

Some HCI methodology has been employed before the development of this project. It includes the characterization of persona, constructing an affinity diagram and the inclusion of mental models for product development.

Persona & User Scenario

Before developing the prototype, user analysis has been conducted to study the potential user of this project. Since the main target of the project would be KSAT takers, the main persona in this project would be Korean Highschooler who are actively reviewing past papers. Some characteristics that were taken into account would be digital proficiency, demographic, geographic and psychological information. The most important characteristics were highlighted as seen in Figure 2.1.1.

Persona

Personal Information



Name: 민준

Demographic information

- High Schooler
- Boy
- No short of getting study materials

Geographic information

- Lives in Mokdong, Seoul
- usually stay at school and study cafe
- hardly move or travel to other region

Psychological information

- after school, moves to study cafe with friends and study individually
- deeply analyze and make feedback after taking mock test
- prefers using Tablet PC rather than pen and notebook, when feedback

Digital proficiency

- uses Phone when not studying
- uses Tablet PC to watch online lectures(인터넷 강의) and to summarize things learned

Figure 2.1.1: Persona

In addition to having a persona, the user scenario acts as an example to show the significance of this project in daily life. An example of a user scenario can be seen in Figure 2.1.2.

User Scenario

[7:00 ~ 8:00]

Wake up and prepare for **school**. Brings essentials such as phones and pens for note taking for most subjects in school.

[8:00 ~ 8:50]

Recap about things learned yesterday (using the app), and a little free time

[8:50 ~ 16:40]

School, have lunch, take classes

[16:40~ 18:00]

Have some free time to do a little bit of studying, might go to cafe with friends if given the opportunity.

[18:00 ~ 18:40]

Dinner before beginning night study sessions

[18:40 ~ 23:00]

Begins studying including:

- watching online lecture using Tablet PC
- taking mock-test, track time (using the app)
- record unsolved questions, self-feedback (using the app)

[23:00~24:00]

Return to home and prepare to sleep

Figure 2.1.2: User Scenario

Affinity Diagram

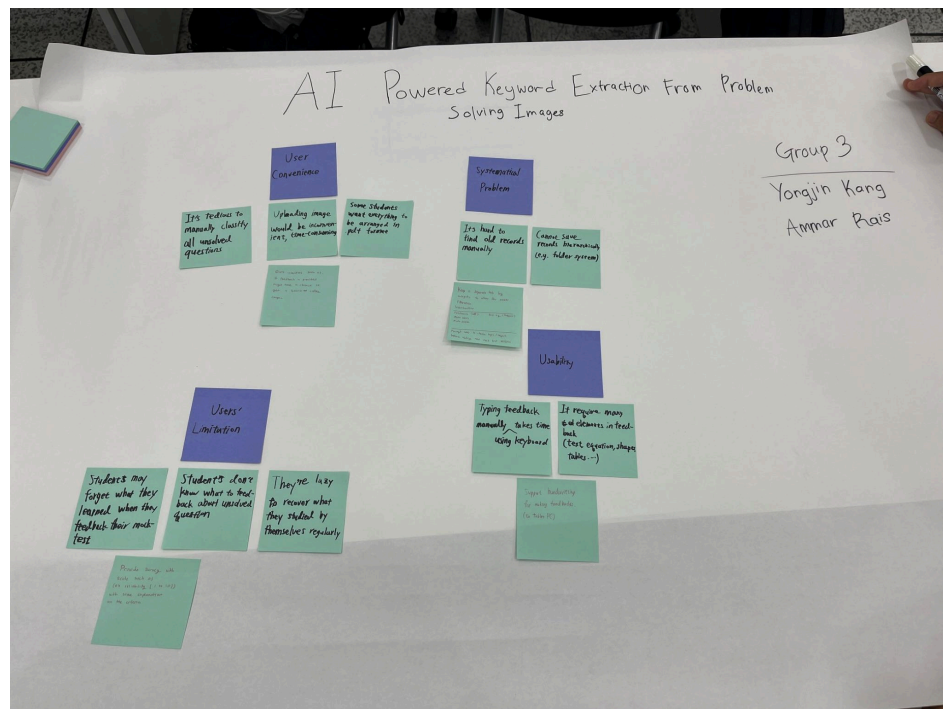


Figure 2.2: Affinity Diagram of Project

After defining target users, the next prospect that can be considered is affinity diagram. Affinity diagrams allow developers to analyze potential problems that may occur throughout the service usage and some recommendations and/or improvements that can be made before the item is fully deployed for user testing. In Figure 2.2, some criteria such as usability and user limitation may produce some problems which were put into consideration and potential solutions for said problems were highlighted.

Mental Model

A mental model was implemented during the development of this project, specifically the functional model, where different color accents are utilized to highlight the importance of extracted keywords as seen in Figure 2.3. As explained in the following section [Lo-fi and Mid-fi prototypes](#), important keywords from the question were ranked by relevance, with distinct colors that become brighter as relevance decreases.

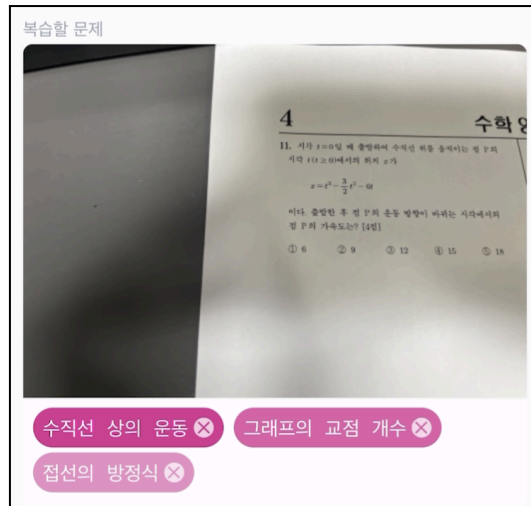


Figure 2.3: Applied mental model through tag colors

Lo-fi and Mid-fi prototypes

The first step of building a prototype was to draw a rough lo-fi prototype of our service. Our main goal was to **build a simple website to test the functionality of LLM-powered keyword extraction, which does not include the whole features of existing app service**. Therefore the lo-fi prototype consists of the required pages and components to test the keyword extraction feature, including input boxes and dropdowns. 4 prototype pages in Figure 3.1 outlines the main workflow of our feature: 1. uploading images, 2. saving extracted keywords and 3. revisiting the saved records.

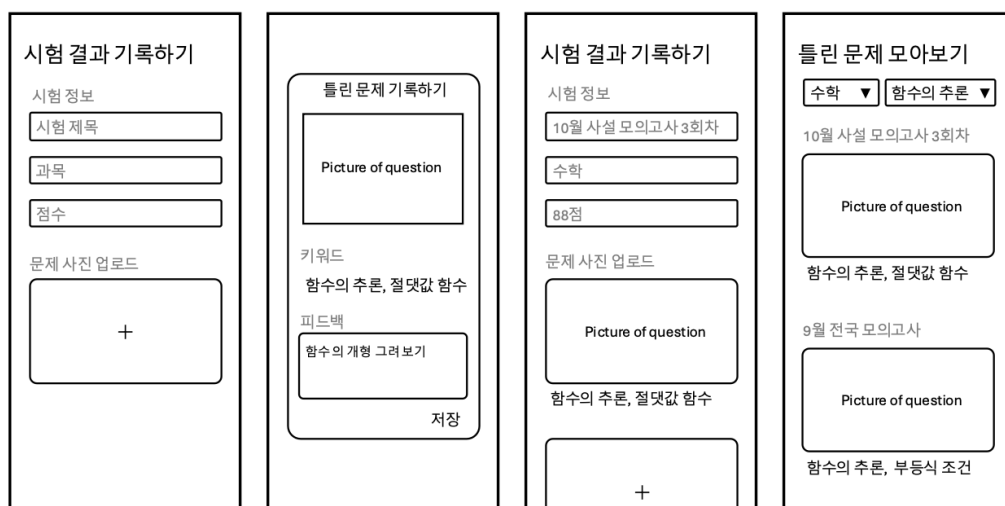


Figure 3.1: Lo-fi prototype

Based on the lo-fi prototype, we developed a mid-fi prototype using **Figma**. It included details of UI designs with color themes, example images that are supposed to be uploaded by the users, and more precise and conditioned workflows based on numerous pages and scenarios. Through developing the mid-fi prototype, we also elaborated more specific functional details like the followings :

- Generated keyword tags should be allowed to be deleted by the user.
- Tags should have distinct colors to emphasize the importance of keywords through varying brightness of colors.

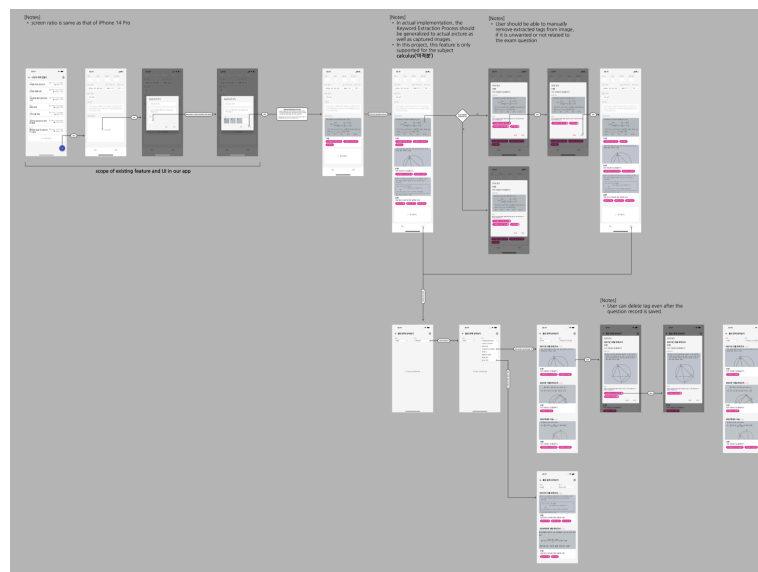


Figure 3.2: Flow of the mid-fi prototype

AI, LLM application

Initially, we considered using a single comprehensive model that takes a question image as input and directly outputs keywords. Thus we considered our task as multi-modal. To address this, we explored models capable of handling both visual and natural language inputs, such as Vision Transformer, and believed we could achieve our goal by fine-tuning a pretrained model.

However, we realized that breaking down the keyword extraction process to two distinct steps would yield better results. (1) First extracting text from the image and (2) selecting top-k keywords relevant to the extracted text. Since the text within the question image encapsulates the entire context of the question, we determined that processing raw images directly to extract keywords was unnecessary. Rather, we anticipated that extracting text using OCR and selecting relevant keywords would be a much more intuitive and effective approach.

Math OCR (Pix2Text)

For OCR, we utilized the Pix2Text python package. This is a comprehensive package that detects Korean, English and mathematical equations at the same time. This allows for a reliable sentence extraction from Mathematics questions, which will be the input for our language model to extract keywords. Following Figure 4.1.1 is an example raw output from Pix2Text, extracted from an image in Figure 4.1.2. Although some characters are misidentified(e.g. ‘를’), important keywords necessary for understanding the context of the question are accurately detected(e.g. ‘수직선’, ‘점 P의 운동’).

“\$11.\$ 시각다 \$t=0\$ 일 때 출발하여 수직선 위를 움직이는 점 P의
시각 \$t \left(t \geq 0 \right)\$에서의 위치 .가
\$\$
x=t^3-\frac{3}{2} t^2-6 t
\$\$
이다: 출발한 후 점 P의 운동 방향이 바뀌는 시각에서의
점 P의 가속도는? [4점]”

Figure 4.1.1

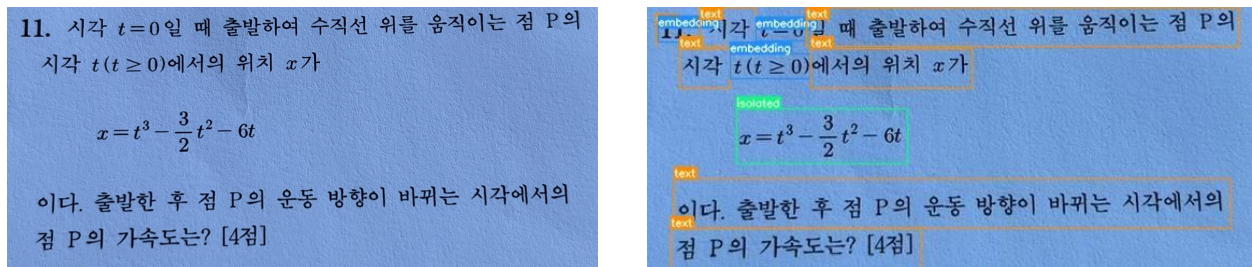


Figure 4.1.2: (Left) input image for OCR, (Right) bounding boxes to identify sentence and equation

One challenge we faced was that OCR took too much time to process images. To resolve this, utilizing GPU power allowed us to process the image faster. Since we ultimately had to run OCR on a cloud server instance, we set up the instance using [Docker image from Nvidia](#). It allowed us to install all drivers regarding CUDA we needed to execute OCR faster.

Language Model

The next step of keyword extraction is inputting the extracted sentences into a language model to identify the top-K keywords relevant to the given question. We first determined that rather than generating keywords using a language model, selecting relevant keywords from a predefined list of possible keywords would be more effective. This decision was based on the fixed scope of KSAT Mathematics, which results in a finite set of possible keywords. Therefore we selected 20 keywords that could be extracted from KSAT Mathematics questions. The selected keywords are as follows:

['구간별로 정의된 함수', '그래프로 둘러싸인 넓이', '그래프의 교점 개수', '등차수열', '로그함수', '미분가능성', '사인법칙과 코사인법칙', '삼각함수', '수열의 귀납적 정의', '수열의 합', '수직선 상의 운동', '외접원과 내접원', '절댓값 함수', '접선의 방정식', '정적분', '정적분으로 정의된 함수', '지수함수', '함수의 극한', '함수의 연속성', '함수의 추론']

Figure 4.2: Potential keywords for math questions in KSAT

After compiling a list of possible keywords, we proceeded to develop a model for the extraction task. Our initial approach was using a pretrained language model such as Llama and fine-tuning it.

Llama-3-Korean Blossom 8B

The backbone model we selected was [llama-3-Korean-Blossom-8B](#), a Korean Llama model with 8B parameters. We froze pretrained parameters and attached a new classifier head to the model as shown in Figure 4.3.1. And we built a dataset consisting of 100 sets of questions in the form of text and corresponding keywords, as presented in Figure 4.3.2. The dataset was used to fine-tune the model.

After fine-tuning the model 30-40 epochs(Figure 4.3.3), the trained model was able to figure out some relevant keywords from images. However the top-K keywords were not consistent, providing non-relevant keywords. Due to time constraints of the project, the model could not be well optimized and fine-tuned.

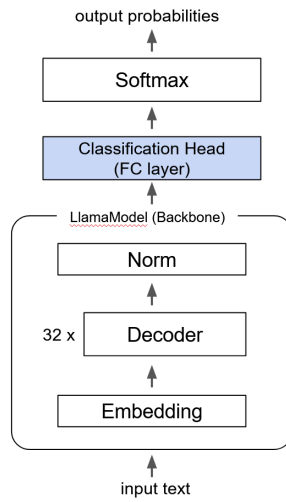


Figure 4.3.1: Structure of fine-tuned model

data_distributed		
from	instruction	output
210609	9. 수직선 위를 움직이는 점 P 의 가속도가 2 m/s^2 일 때 시간 $t=0$ 부터 $t=10$ 까지 점 P 가 움직인 거리는? (단, S_0 는 상수이다.) (4점) S_0 $v(t) = -4t^2 + 12t + 10$ S_0 이다. 시간 $t=0$ 에서 점 P 의 가속도가 2 m/s^2 일 때 시간 $t=0$ 에서 $t=10$ 까지 점 P 가 움직인 거리는? (단, S_0 는 상수이다.) (4점)	수직선 상의 운동, 정적분
210910	10. 두 양수 a, b 에 대하여 극한 $\lim_{n \rightarrow \infty} \frac{a^n + b^n}{a^n + b^n} = \frac{a}{b}$ 이라 하자. 이 때 a, b 의 값은? (단, a, b 는 양수이다.) (4점)	삼각함수
210911	11. 다항함수 $f(x)$ 가 모든 실수 x 에 대하여 $f(x) = 2x^2 + 3x + 1$ 이라 하자. 이 때 $f(1)$ 의 값은? (단, S_0 는 상수이다.) (4점)	함수의 추론, 정적분
210912	12. 만지불러 값이 0.2 인 $\log_{10}(x)$ 를 x 에 대입하고 $\frac{1}{x}$ 를 대입하면 $\frac{1}{x}$ 의 값은? (단, S_0 는 상수이다.) (4점) $\log_{10}(x) = 0.2$ 이라 하자. 이 때 $\frac{1}{x}$ 의 값은? (단, S_0 는 상수이다.) (4점)	사건법칙과 공산법칙, 국소화와 내림법

Figure 4.3.2: Dataset used for fine-tuning

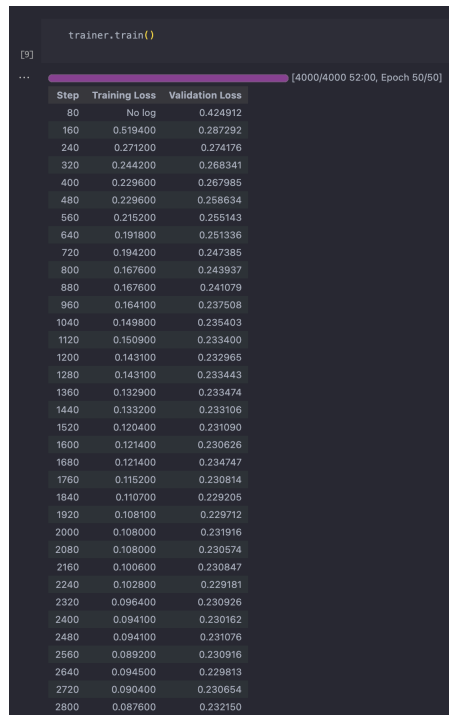


Figure 4.3.3: Training process of the pre-trained model

Sentence Transformer

Instead of fine-tuning a pre-trained keyword extractor, a model named **Sentence Transformer** also known as **SBERT** was introduced to implement a zero-shot solution. SBERT works by performing embedding on input (in this case text-type input), and finding the semantic similarity between the embeddings of questions (raw text input from OCR) and keywords (20 possible mathematical keywords). This removes the need to fine-tune our own model, while providing relevant keywords at top-K (outperforming pre-trained Llama model). However, there are some limitations to the model, one of them would be incorrect keywords predicted. This might cause issues if the user did not manually remove/edit the keywords which may not have any relevance to the question. One plausible solution for this would be to set a lower K for top-K keywords, though it may not be significant for some difficult questions that may have many relevant keywords.

```
Given question:
$11.$ 시각다 $t=0$ 일 때 출발하여 수직선 위를 움직이는 점 P의
ight)$ 에서의 위치 .가
$$
x=t^{\{3\}}-rac{\{3\} \{2\} \, , \, t^{\{2\}}-6 \, t
$$
이다: 출발한 후 점 P 의 운동 방향이 바뀌는 시각에서의
점 P 의 가속도는? [4점]

Top Keywords: ['수직선 상의 운동', '그래프의 교점 개수', '함수의 추론']
Similarities: [0.37630748748779297, 0.1790696382522583, 0.1396966278553009]
```

Figure 4.4: Top 3 keywords with similarities score, using example from Figure 4.1.1

Development

Back-end

Instead of running the whole keyword extraction process directly on-device, we chose to run the keyword extraction process on the separated server with GPU capabilities. Then we implemented an API which takes a question image as input and returns extracted keywords. The API was used at frontend to communicate with the server. The server providing the API was set up using **FastAPI**.

We also had to address the 10-seconds timeout limit for every API usage imposed by [Vercel](#), the web deployment service we used. Even though we utilized GPU to speed up the OCR process by setting up the instance using Docker image from Nvidia, it required a server instance with a decent CPU and GPU. If the server instance is built on an instance with a limited computing resource, it took more than 10 seconds to extract keywords, resulting in a 504 Gateway timeout error at the frontend

side. Therefore, we built our server on a cloud instance provided by vast.ai, which runs in a Python environment. This server environment setup allowed us to utilize better GPU resources for faster OCR processing. In the instance, we set up all modules including Pix2Text and Sentence Transformer into a server. The server was operated during the day (9am-10pm) throughout the survey period.

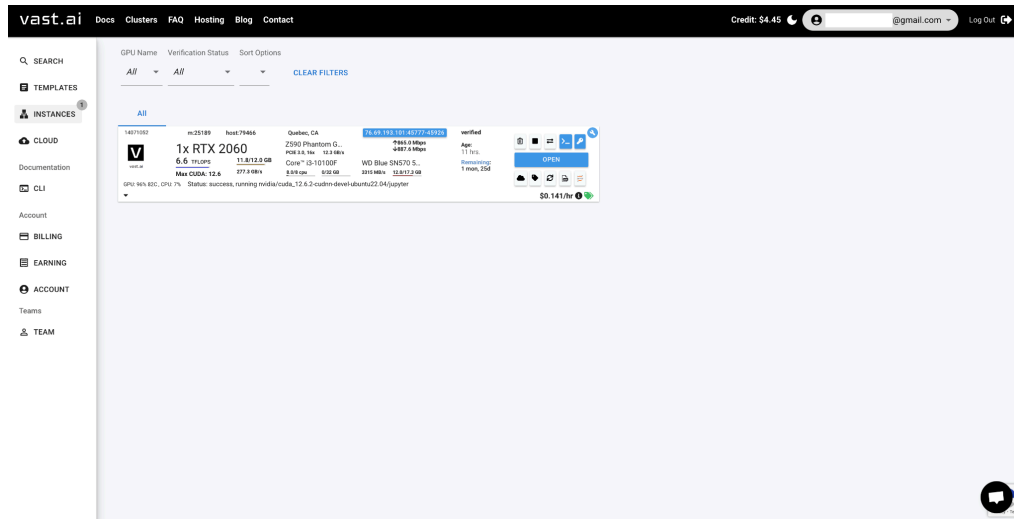


Figure 5: GPU rental for testing LLM

Front-end

To develop the frontend UI for our Minimum Viable Product (prototype), we prioritized (1) rapid development and deployment, and (2) a responsive way to test the prototype on both mobile phones and tablet PCs. To meet these requirements, we utilized the **Next.js** web development framework to build a website prototype. Next.js enables easy and rapid deployment through the web publishing service [Vercel](https://vercel.com). Additionally, by developing a responsive website, we ensured compatibility across both mobile phones and tablet PCs.

The code for the frontend of the prototype is available on [Github](https://github.com). The prototype is published [here](#) and still running, but keyword extraction is currently unavailable as the backend server instance has been turned off.

Survey

After developing the prototype, we desired it to be tested and evaluated by actual users. For this purpose, we recruited actual users of the existing app service who were willing to participate in testing the prototype. When providing access to the prototype website, we also collected **pre- and**

post-survey responses. By analyzing the response data with a paired t-test, we aimed to measure users' satisfaction with the new feature.

Questionnaires

The questionnaire in each survey consisted of 4 questions in 5 point likert scale as reported in Table 1. Among the questions, Q1, Q3 aimed to measure **improvement of willingness to use the app service to record and revisit unsolved questions**, while Q2, Q4 aimed to measure **improvement of users' study habits**.

	Before	After
Q1	나는 실전 모의고사 응시 후 틀린 문제/시간 내에 풀지 못한 문제에 대해 실감 앱으로 오답노트를 작성한다.	나는 앞으로 실전 모의고사 응시 후 틀린 문제/시간 내에 풀지 못한 문제에 대해 실감 앱으로 오답노트를 작성할 것이다.
Q2	나는 실전 모의고사 응시 후 틀린 문제/시간 내에 풀지 못한 문제를 풀기 위해 필요했던 핵심 아이디어가 무엇인지 고민해본다.	나는 앞으로 실전 모의고사 응시 후 틀린 문제/시간 내에 풀지 못한 문제를 풀기 위해 필요했던 핵심 아이디어가 무엇인지 고민해볼 것이다.
Q3	나는 실전 모의고사 응시 후 앱에 기록했던 오답 문항들을 이후 다시 복습한다.	나는 앞으로 실전 모의고사 응시 후 앱에 기록했던 오답 문항들을 향후 다시 복습할 것이다.
Q4	나는 수학 과목의 실전 모의고사에서 틀렸던 문제들을 주제별로 모아서 복습한다.	나는 앞으로 수학 과목의 실전 모의고사에서 틀렸던 문제들을 주제별로 모아서 복습할 것이다.

Table 1: Question asked during survey (left) before and (right) after

Recruitment of participants

As for test participants, we hoped to recruit students who prepared this year's KSAT. For this purpose, we started posting a survey banner in our app service 2 weeks before the KSAT(November 14, 2024) as seen in Figure 6.1. The banner was redirected to a Google form including a question asking users to participate in a test of our prototype later in this year. After recruiting potential participants in early-November, we conducted an actual testing and survey after completion of building our prototype, which started from November 24, 2024.

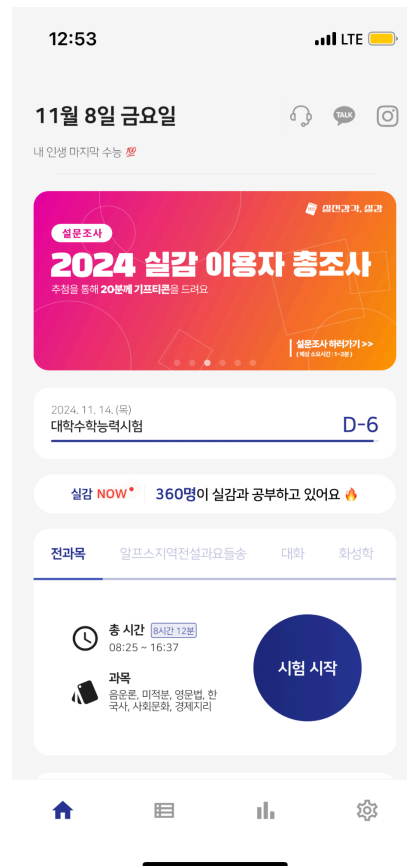


Figure 6.1: Survey banner being posted on the existing app

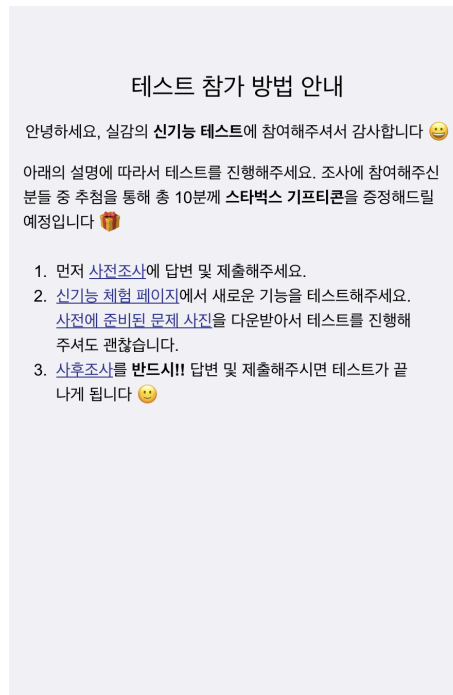


Figure 6.2: Instruction page given to participant

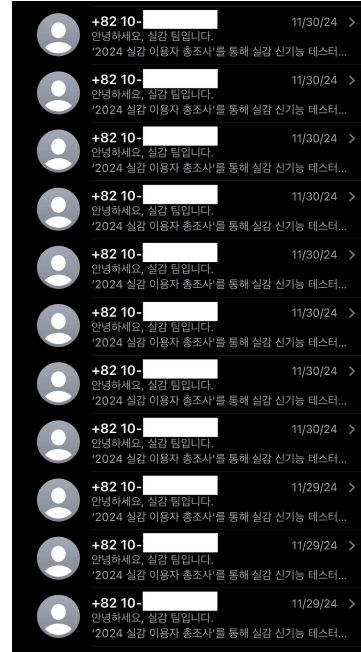
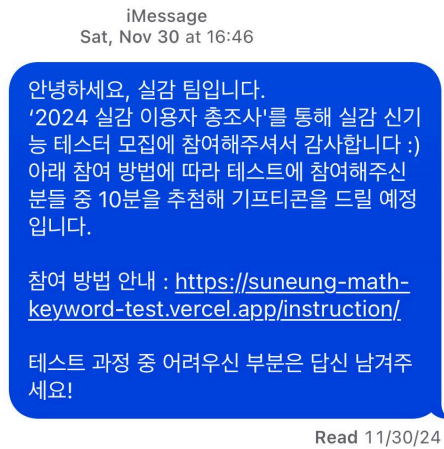


Figure 6.3: Message sent to participant

Before starting the test, we built an [instruction page](#) (Figure 6.2) to guide users to respond to 2 surveys before and after they test the prototype. And we sent an url to the instruction page to recruit participants through SMS on November 24, 2024 ~ December 4, 2024. We sent **292** messages in total and collected **29** responses, with approximately 10% of response rate.

Analysis

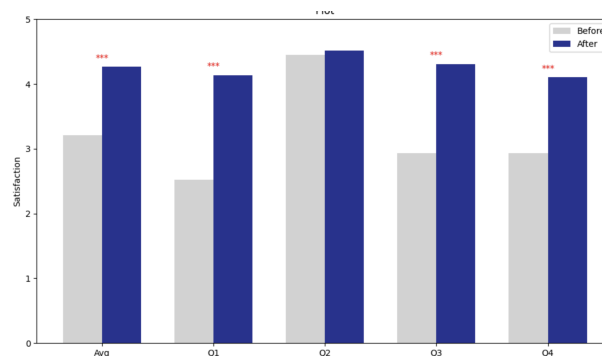


Figure 7: Improvement on mean and each question (before and after)

We performed a **paired t-test** on the 29 collected responses. The mean satisfaction of Q1-Q4 improved after users tried the prototype, and the improvement was statistically significant

($t(28)=-6.62$, $p<0.001$). Furthermore, improvement of satisfaction for each question was individually significant at the 0.1% confidence level, except for the Q2.

Statistically significant improvement of Q1 and Q3 indicates that our prototype had an impact on enhancing users to use our app service more, for the sake of recording unsolved questions. The statistically significant improvement of Q4 indicates that our prototype effectively encourages users to revisit recorded questions, leading to an enhancement in their study habits.

However, unlike our expectation, insignificance of improvement in Q2 means that people have already been analyzing key ideas of unsolved questions, after solving mock tests. Thus our prototype did not show a meaningful impact on this aspect.

Future Work

OCR (Improper result from non-ideal input image)

There are some limitations with the OCR that was used (Pix2Text), mainly the lack of ability to extract text from images with little text and more numbers as seen in Figure 8.1.1 and 8.1.2. Another problem from the OCR usage includes text extraction from handwritten texts on test papers as seen in Figure 8.1.3. Ideally, to get the best possible text extraction, the user needs to ensure that the problem image only includes part of the question (answers option, user handwriting, and figures such as graphs or charts should be excluded) as shown in [Math OCR\(Pix2Text\)](#). One possible solution for this would be to fine-tune the OCR since the native support for Pix2Text is Chinese letters with multilingual support such as Korean letters.

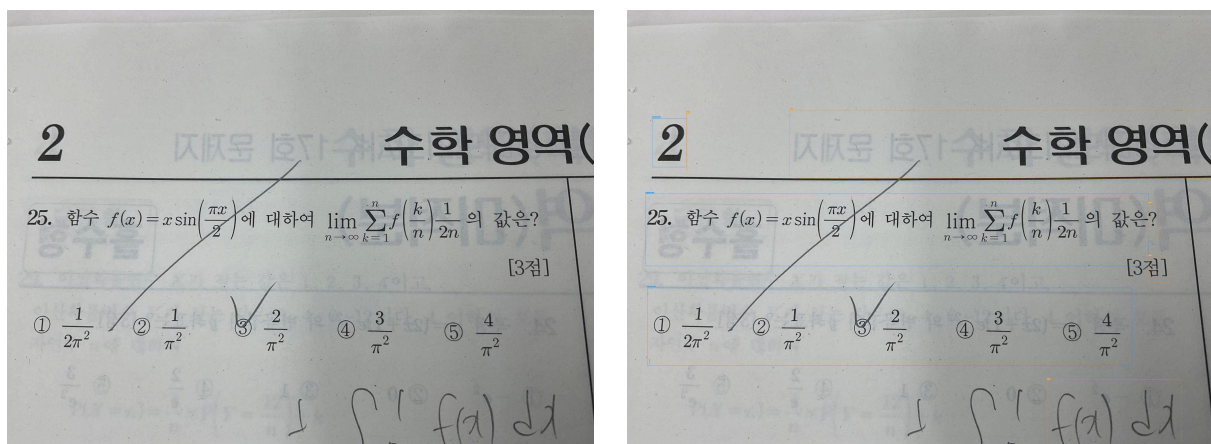


Figure 8.1.1: (Left) input image for OCR, (Right) bounding boxes to identify sentence and equation

$\int_0^{\pi} \sin(x) dx = -\cos(x) \Big|_0^{\pi} = -\cos(\pi) + \cos(0) = -(-1) + 1 = 2$

점]

$\frac{1}{\pi^2}$; $\frac{1}{\pi^2}$; $\frac{1}{\pi^2}$; $\frac{1}{\pi^2}$; $\frac{1}{\pi^2}$; $\frac{1}{\pi^2}$; $\frac{1}{\pi^2}$; $\frac{1}{\pi^2}$; $\frac{1}{\pi^2}$; $\frac{1}{\pi^2}$

Figure 8.1.2: Extracted text from question in Figure 8.1.1

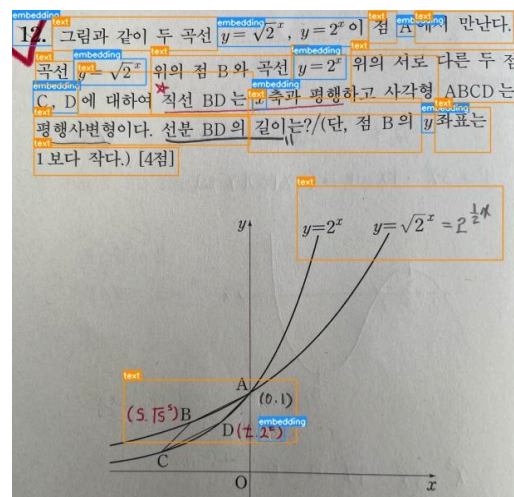


Figure 8.1.3: Text extraction from user handwriting

Fine tuning

Instead of relying on SBERT for a zero-shot solution, we could try to improve the keyword extraction through a fine-tuned model mentioned in [Llama-3-Korean Blossom 8B](#) to get a better similarity score (similarity > 0.5) which would reduce the likelihood of suggesting irrelevant keywords for question.

Expansion toward different subjects and examinations

Since our prototype only works on the Mathematics subject of KSAT, it is not usable on other subjects that are present in KSAT. In terms of composition of exam papers, Mathematics is not the same as Language(국어), History(한국사) or any other subjects. To expand toward such subjects, some potential challenges that may appear for subjects such as Language(국어) would be the

processing of much longer text input and summarizing them in a few keywords. Furthermore, for subject of Social study(사회 탐구), it would require the model to process various types of inputs (e.g. tables or graphs), which is not capable in our prototype. But by overcoming such difficulties, ideally our prototype also can be expanded on more difficult exams such as TOEIC, ACCA, Bar Exam, etc.

On-device implementation of the AI models

One feedback from users in the post-survey was that waiting 6-7 seconds for the keyword extraction process was inconvenient. Thus another potential implementation that can be done for this work is the usage of an on-device application of the AI models. It has potential to allow faster execution by eliminating the need for transactions with a server instance. However, this would be challenging since the models must be optimized to run on mobile phones and tablet PCs efficiently.

Discussion

Through the whole process of developing the prototype and taking user study, we figured out that our new feature of extracting keywords from question images has potential to help students study better. Moreover, it is one example that utilizing AI-powered service as an aid for students is a relevant way to help students nowadays. However, before we introduce AI more widely in education, there are threats to be more carefully considered to ensure that the service does not harm students. One example of this is jailbreaking, in which we must prevent the model from providing wrong/harmful answers. Taking the considerations into account, we can look forward to a better future of enhanced education with AI.

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